Hospital Readmission & Patient Outcome Analysis

Power BI Insights & Management Summary

CITY CARE HOSPITAL

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Analysis Questions & Power BI Insights

Six guiding questions were used to explore readmission drivers, cost efficiency, and patient risk factors across City Care Hospital's departments.

1,000	7.46 days	₦330.28K	21.0%
Total Patients	Average Length of Stay	Average Treatment Cost	Overall 30-Day Readmission Rate

Q1 Which month has the highest readmission rate?

KEY FINDING — October recorded the highest 30-day readmission rate at 26.7%.

The higher readmission rate in October indicates that patients discharged during this month were more likely to return within 30 days than in any other month. This may suggest seasonal pressures, variations in patient case complexity, discharge practices, or resource availability that affected patient outcomes.

MANAGEMENT ACTION

Hospital management should investigate the factors contributing to the high readmission rate in October by reviewing discharge procedures, staffing levels, patient case mix, and post-discharge follow-up. Identifying the underlying causes can help reduce avoidable readmissions and improve patient care.

Q2 Which department contributes most to readmissions?

KEY FINDING — The Cardiology department recorded the highest 30-day readmission rate at 27.7%.

The higher readmission rate in the Cardiology department suggests that patients treated for cardiovascular conditions are more likely to be readmitted within 30 days than patients in other departments. This may reflect the complexity of cardiac conditions, the need for ongoing treatment, or challenges in post-discharge care.

MANAGEMENT ACTION

Hospital management should review discharge protocols and follow-up care within the Cardiology department. Strengthening patient education, medication management, and post-discharge monitoring may help reduce avoidable readmissions and improve patient outcomes.

Q3 Do shorter hospital stays result in higher readmissions?

KEY FINDING — No clear evidence that shorter hospital stays result in higher readmissions.

Based on the dashboard, there is no clear evidence that shorter hospital stays result in higher readmissions. While the Cardiology department has a relatively short average length of stay (7.20 days) and the highest readmission rate (27.7%), other departments with similarly short stays, such as Pediatrics (7.04 days), have much lower readmission rates (18.3%).

The findings suggest that length of stay alone does not explain readmission rates. Other factors such as the severity of patients' conditions, quality of treatment, discharge planning, and follow-up care are likely to have a greater influence on whether patients are readmitted.

MANAGEMENT ACTION

Hospital management should avoid using length of stay as the sole indicator of readmission risk. Instead, they should investigate additional clinical and operational factors, particularly in departments with high readmission rates, to identify opportunities for improving patient outcomes.

Q4 Which age group is most at risk?

KEY FINDING — Patients aged 61 and above have the highest readmission rate, at 37.7%.

The significantly higher readmission rate among patients aged 61 and above suggests that older adults are more vulnerable to complications or require additional care after discharge. This indicates that age may be an important factor influencing the likelihood of readmission.

MANAGEMENT ACTION

Hospital management should implement targeted interventions for elderly patients, such as enhanced discharge planning, medication reviews, follow-up appointments, and post-discharge support. These measures can help reduce readmissions and improve patient outcomes for this high-risk age group.

Q5 Are higher treatment costs linked to better patient outcomes?

KEY FINDING — No clear evidence — the Deceased category had the highest average treatment cost (₦385.88K).

Based on the dashboard, there is no clear evidence that higher treatment costs are linked to better patient outcomes. In fact, patients in the Deceased category had the highest average treatment cost (₦385.88K), while the Recovered and Improved groups had lower average treatment costs.

The findings suggest that higher treatment costs do not necessarily result in better patient outcomes. Higher costs may instead reflect the complexity or severity of a patient's condition, which can require more intensive treatment without guaranteeing recovery.

MANAGEMENT ACTION

Hospital management should evaluate the factors driving high treatment costs, particularly for patients with poor outcomes. Reviewing treatment pathways, resource utilization, and clinical practices may help improve both cost efficiency and patient outcomes.

Q6 Which departments are high cost but poor outcome?

KEY FINDING — ICU, Oncology, and Cardiology combine high treatment costs with high readmission rates.

The ICU, Oncology, and Cardiology departments stand out as having relatively high treatment costs while also recording high readmission rates. ICU had the highest average treatment cost (₦627,742.96) with a 23.5% readmission rate, Oncology recorded an average treatment cost of ₦526,869.56 with a 24.4% readmission rate, and Cardiology had an average treatment cost of ₦378,817.99 alongside the highest readmission rate of 27.7%.

These findings suggest that higher spending in these departments is not necessarily associated with lower readmission rates. The combination of high treatment costs and relatively high readmissions may reflect the complexity of the conditions being treated or indicate opportunities to improve care pathways and post-discharge support.

MANAGEMENT ACTION

Hospital management should prioritize a detailed review of the ICU, Oncology, and Cardiology departments to identify the factors driving both high treatment costs and high readmission rates. Evaluating clinical workflows, discharge planning, and follow-up care may help improve patient outcomes while making better use of hospital resources.

Management Summary

The Hospital Readmission & Patient Outcome Dashboard was developed to evaluate patient admissions, 30-day readmission rates, treatment costs, length of stay, and patient outcomes across different hospital departments. The analysis provides valuable insights into areas requiring improvement and supports data-driven decision-making to enhance patient care and operational efficiency.

The dashboard revealed an overall 30-day readmission rate of 21.0% among 1,000 patients. October recorded the highest monthly readmission rate at 26.7%, indicating that patients discharged during this period were more likely to be readmitted than in other months. At the departmental level, Cardiology had the highest readmission rate (27.7%), followed by Oncology (24.4%) and ICU (23.5%), suggesting that these departments require closer monitoring and improved discharge planning.

The analysis also identified patients aged 61 years and above as the highest-risk group, with a readmission rate of 37.7%, considerably higher than all other age groups. This indicates the need for enhanced post-discharge care, medication management, and follow-up support for elderly patients. Additionally, there was no clear evidence that shorter hospital stays resulted in higher readmission rates, suggesting that factors beyond length of stay such as patient condition and quality of discharge planning play a more significant role in determining readmission risk.

From a financial perspective, higher treatment costs did not consistently correspond to better patient outcomes. Patients in the Deceased category recorded the highest average treatment cost, indicating that increased healthcare spending may reflect the complexity or severity of cases rather than improved outcomes. Furthermore, the ICU, Oncology, and Cardiology departments combined relatively high treatment costs with comparatively high readmission rates, highlighting opportunities to improve both cost efficiency and quality of care.

Based on these findings, hospital management should prioritize strengthening discharge planning and follow-up care in the Cardiology, Oncology, and ICU departments, with particular attention to patients aged 61 years and above. Investigating the causes of the increased readmission rate observed in October, reviewing treatment pathways in high-cost departments, and continuously monitoring key performance indicators through interactive dashboards will support better patient outcomes, reduce avoidable readmissions, and improve resource utilization across the hospital.